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RESEARCH ARTICLE



A comparison of video modeling and video-animated modeling in teaching social-affective behaviors to children with ASD

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ABSTRACT

Individuals with autism spectrum disorder experience difficulties in the social interaction and communication domain, which includes recognising and expressing emotions, as well as observable behaviours such as verbal and facial expressions, gestures, and intonations, also known as social-affective behaviours. Such individuals need to learn social-affective behaviours appropriate for the social context to perceive the reasons and conditions that give rise to emotions. This study aimed to compare the effectiveness and efficiency of video modelling (VM) and video-animated modelling (VAM) in acquiring, maintaining, and generalising three social-affective behaviours (verbal and facial expressions and gestures) together in children with autism spectrum disorder. Three children aged between 4 and 6 years with autism spectrum disorder and their mothers participated in the study. The study used the nested form of a single-case design with non-concurrent multiple baselines and an adapted alternation of VM and VAM conditions. The results indicated that both VM and VAM procedures were effective, with no consistent or systematic differences across participants to acquire contextually appropriate social-affective behaviours including verbal and facial expressions and gestures. The efficiency findings varied across participants in terms of the number of errors and total time. Social validity findings showed that the participating children's mothers and teachers expressed positive opinions on the interventions. Recommendations and future research issues are discussed in line with the findings.

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Autism spectrum disorder; video modelling; video-animated modelling; social-affective behaviours; verbal expression; facial expression; gesture

► IMPLICATIONS FOR REHABILITATION

Social-affective behaviours refer to observable behaviours, such as verbal and facial expressions, gestures, and intonations, displayed to recognise and express emotions with a social communicative function. Children with autism spectrum disorder (ASD) have difficulties in social interaction and communication manifested as an inability to understand and express emotions through social-affective behaviours. They display social-affective behaviours less accurately, and in out-of-context. Video-animated modelling is as effective as video modelling in teaching social-affective behaviours to children with ASD. Therefore, teachers can prefer to use video animated modelling, as an alternative and enjoyable option, in teaching social-affective behaviours appropriate for the social context to perceive the conditions that give rise to emotions.

Social interaction requires the ability to recognise and express emotions within social contexts. When individuals recognise emotions, understand the behaviours that convey these emotions, and adjust their own behaviour accordingly, social interaction becomes appropriate and sustainable [1,2]. Appropriately responding to discriminative stimuli in social contexts and to the affective behaviours of others enables connections among contexts, affective states, and the messages conveyed through social-affective behaviours [2,3].

Social-affective behaviours (SABs) are critical for individuals to share joint experiences, discriminate social cues, and modify their behaviour in response to social demands through both verbal and nonverbal means [2,4]. SABs refer to observable behaviours—such as verbal and facial expressions, gestures,

and intonation—that serve to recognise and express emotions within a social-communicative function [2,3,5,6]. Individuals with autism spectrum disorder (ASD) experience difficulties in social-affective functioning, manifested as reduced attention to social stimuli, challenges in understanding and expressing emotions, and difficulty adjusting behaviour across diverse social contexts [7]. These difficulties negatively affect overall social development and limit interpersonal interactions [6,8].

Verbal expressions, as one component of SABs, serve as a fundamental channel for communication, enabling individuals to express emotions appropriately within social contexts [9]. Children with ASD are unable or limited in their use of language to respond to social situations and share their emotions [10,11].

Facial expressions and *gestures* are among the most important social-affective behaviours for recognising emotions and rendering social interactions meaningful [12]. Children with ASD often demonstrate limited spontaneous abilities to recognise, imitate, use, and understand facial expressions and gestures [12,13]. When social-affective behaviours are considered collectively, children with ASD display these behaviours less frequently, less accurately, and in less contextually appropriate ways than their typically developing peers [14,15]. Accordingly, explicit instruction in contextually appropriate social-affective behaviours is needed to help these children understand the conditions and causes underlying emotional experiences.

In social interactions, verbal expressions, facial expressions, and gestures typically do not occur as independent units but rather emerge in a complementary and synchronous manner. Therefore, teaching SAB components together may support individuals with ASD in coordinating these components simultaneously and in producing appropriate social responses [16,17]. Numerous studies have taught SABs individually to individuals with ASD (e.g., [18–21]). In contrast, relatively few studies have examined the simultaneous teaching of two or more SABs [1–4,22,23]. These studies have employed a range of instructional procedures—including video modelling (VM), *in vivo* modelling, script fading, prompting, reinforcement, and behavioural rehearsal—to teach appropriate affective responding to individuals with ASD.

Video modelling (VM), an evidence-based practice, involves the observation of a modelled behaviour presented in a video and the subsequent performance of the same or a similar behaviour by the learner [24]. A substantial body of research has demonstrated the effectiveness of VM [25–28]; however, relatively few studies have used VM to teach SABs individually [29–34]. Moreover, only a limited number of studies have targeted the simultaneous instruction of two or three SABs using VM [2,3]. Gena et al. [2] compared VM and *in vivo* modelling in teaching appropriate verbal and facial responses as affective behaviours to pre-school children with ASD and reported increases in affective responding across all response categories for three participants. Similarly, Pektas-Karabekir and Akmanoglu (2018) used VM to teach verbal expressions, facial expressions, and gestures to three children with ASD, demonstrating that VM delivered *via* a smartboard facilitated the acquisition and generalisation of these social responses across different settings, materials, and individuals. Despite these findings, only one study has attempted to teach the full range of SABs using VM [4]. Charlop et al. [4] examined the effectiveness of VM in teaching verbal expressions, intonations, gestures, and facial expressions concurrently to children with ASD during social interactions. Results indicated that VM promoted rapid acquisition and generalisation of SABs across people, settings, and materials. However, the study did not explicitly incorporate the affective states associated with social responses or the contextual conditions under which emotions emerge.

Teaching SABs is a complex process, and video-animated modelling (VAM) has been proposed as an alternative instructional approach for teaching SABs [35,36]. In VAM, an animated model is presented *via* video, depicting a character aligned with the child's special interests while performing the targeted skill or behaviour, and the video is then shown to the child [37]. The use of animations in education contributes to making abstract knowledge and skills more concrete through visual content and to enhancing the retention of acquired knowledge and skills [38]. In the education of children with ASD, the use of animations supports active participation in activities and promotes learning through enjoyment. In particular, more effective outcomes are achieved when animations are designed in accordance with children's interests, needs, attention span, and developmental characteristics [39]. Research comparing animated and human models indicates that children with ASD often respond

more positively to animated models than to human models [40,41]. To date, a limited number of studies have examined the effectiveness of instructional practices using VAM, and these studies have primarily focused on teaching self-care and academic skills to children with ASD [37,42–44]. To the best of our knowledge, no study has investigated the effectiveness of VAM in teaching social-affective behaviours.

Many studies have compared VM with different teaching practices in terms of effectiveness and efficiency (e.g., [45,46]). However, only one study compared VM with VAM, which has been on the agenda more recently [41]. Li et al. [41] compared the effectiveness of VM, VAM, and mixed VM (VM+VAM) using an adult model in teaching social communication skills to children with ASD. Two three-year-old boys and one girl with ASD participated in the study. The findings showed that VAM was more effective than VM; for two of the three children, mixed VM was more effective than VM and VAM alone. To our knowledge, no research has compared the effectiveness of VM and VAM in teaching three of the SABs (verbal and facial expressions and gestures). Although VAM procedures may offer certain advantages, their development is a more time-consuming and resource-intensive process compared to VM. Therefore, it is an important research question whether these additional costs and efforts lead to superior learning outcomes or whether similar results can be achieved with VM, which requires fewer resources. The current study aimed to compare the effectiveness and efficiency of VM and VAM in teaching three SABs together to children with ASD, including verbal and facial expressions and gestures. In this study, effectiveness refers to the acquisition, maintenance, and generalisation of SABs, whereas efficiency refers to variables such as the number of sessions, number of trials, number of errors, and total instructional time. To this end, answers to the following research questions (RQ) were sought:

RQ1. Does the effectiveness of VM and VAM in teaching SABs in the form of verbal expressions, gestures, and facial expressions, indicative of the targeted affective states, to children with ASD differ at the acquisition, maintenance, and generalisation levels?

RQ2. Do VM and VAM differ in terms of the number of sessions, trials, errors, and total instructional time until the criterion is met in acquiring the targeted SABs?

RQ3. Do the effect size values of VM and VAM differ in terms of Tau-U effect size in acquiring three of the SABs?

RQ4. What are the opinions of the participating children, their mothers, and teachers on the study's purpose, practices, and findings?

Method

Participants

The child participants in the study were three boys (Emir, Aslan, and Kerem), aged between 4 and 6, who had previously been diagnosed with atypical autism by a child psychiatrist at a hospital. In accordance with current diagnostic classification systems (DSM-5; ICD-11), the participants were assessed as having ASD. All children benefit from full or part-time education in a public kindergarten and receive education from a special education centre for four or six hours a week. Table 1 lists the general characteristics of the participating children. Prior to the study, the Gilliam Autistic Disorder Rating Scale-2-Turkish Version (GARS-2-TV) [47], Gazi Early Childhood Development Assessment Tool (GECDAT) [48], Test of Early Language Development-Third Edition (TELD-3) [49], and Stanford-Binet Intelligence Test-Turkish Version [50] were administered to all participating children (Please see details in Table 1).

The following prerequisite characteristics were sought for children to take part in the study: a) understanding and following instructions that state at least two actions, b) using sentences with at least two words, c) imitating verbal expressions with at least two words, d) directing attention to the computer screen and watching the video for at least two-three minutes, e) participating in the activity and directing attention for at least five minutes, f) being able to take turns, g) making eye contact, and h) imitating the model's facial expressions and motor movements. Whether the children had the prerequisite skills was determined through interviews with their mothers and a subsequent pre-assessment session.

Table 1. Demographic information on participants.

Name	Gender	Age	Diagnosis	GARS-2-TV	GECDAT	TELD-3		Standford-Binnet
						RL	EL	
Emir	Male	5 years and 8 months	Atypical autism	83	PMD: 73 CD: 60 LD: 58 SED: 56	80	78	80
Aslan	Male	4 years and 6 months	Atypical autism	83	PMD: 70 CD: 61 LD: 57 SED: 51	114	88	95
Kerem	Male	5 years and 8 months	Atypical autism	97	PMD:71 CD:48 LD:49 SED:53	58	57	67

Note. CD: Cognitive development; EL: Expressive language; LD: Language development; PMD: Psycho-motor development; RL: Receptive language; SED: Social emotional development.

Three mothers participated in the study as their children's play partners. Mercan, one of the participating mothers, is 34 years old and has two children. She is a homemaker and is currently pursuing an associate degree. Leyla is 44 years old and has two children. She is a high school graduate and a homemaker. Maide is 45 years old and has two children. She is a primary school graduate and a homemaker. According to information provided by the mothers, they noted that their children engaged in more interactive play when they guided them. However, they also noted that their children did not exhibit SABs during these play sessions. A session was organised that included verbal explanations and practical demonstrations regarding what was expected of the mothers, such as playing games with their children using scenarios. During these sessions, the mothers were informed about the practical steps involved, such as presenting target stimuli, providing attention-grabbing cues and presenting discriminative stimuli. The reliability of the implementation was monitored using a checklist prepared by the researchers.

Models

In VM intervention sessions, a 10-year-old TD peer model who received theatre training was included as a model for children with ASD, and a research assistant who completed her Ph.D. in special education, as the peer model's mother, was included as an adult model. The models were informed about the research process, the importance of their participation in this study, and what was expected from them as models.

Implementer and observers

The study's implementer is the article's first author. The implementer is pursuing her education in the Doctoral Program in the Education of Individuals with Intellectual Disabilities in the Department of Special Education at a university and has experience in providing one-to-one training to children with ASD. The second author was actively involved in the planning and reporting of this study, with more than 20 years of experience in the field of special education as a trainer of trainers and one-to-one and group training.

Two observers who have doctoral degrees in special education collected data on treatment fidelity and inter-observer reliability. The observers were trained by the implementer on the procedure.

Settings

Intervention, probe, and maintenance sessions for VM and VAM were conducted in the living room of the participants' homes, while generalisation sessions were held in different settings for each participant, including the hallway, kitchen, and children's room. There were generally single and triple armchairs, a coffee table, a TV unit, a toy basket, and a carpet on the floor in the settings for the sessions, other than generalisation sessions. The sessions took place in spaces where the children played with their mothers. During the sessions, the mother and child sat side by side on a couch, and videos were shown to them on a tablet while they were in this position.

Materials

The study used the following materials: a) scenarios, b) videos with peer models, c) videos with animated models, d) materials used in the scenarios (Please see details in [Table 2](#)), e) a tablet, f) data collection forms, and g) reinforcers. Identical materials were used in all baseline, probe and intervention sessions, with the exception of the reinforcers. For social scenarios, the relevant studies in the literature were reviewed, and a scenario pool consisting of 26 social scenarios was created by interviewing four pre-school teachers and three mothers of children with ASD. Four experts working in special education were asked to express their opinions on the scenarios' appropriateness. In line with the expert opinions and the evaluation made with TD children and children with ASD, eight social scenarios were determined for intervention sessions and four social scenarios for generalisation sessions, each including a different emotion. [Table 3](#) lists the social scenarios used to teach each emotion.

While preparing the video images of the peer and adult models, the following conditions were considered: the peer model's gestures and facial expressions should be clearly visible, attention was paid to sound quality, distracting stimuli were removed from the setting, and the model appropriately reflected emotions, intonations, and gestures. Then, three experts working in the field of special education were consulted to evaluate the video images' validity. In line with the experts' opinions, all video images were finalised. These images were then shared with the relevant company to prepare animations. The video clips featuring animated characters were created in two dimensions (2D), considering the characteristics of children with OSB (Please see details in [Supplemental Material 1](#)). VAM is conceptualised as an animated modelling format that is not necessarily customised to individual interests.

The animated videos were developed in collaboration with a professional team. The animations used in the videos were created based on the characters of the models presented in the VM videos. Accordingly, the animated videos included a total of four scenarios consisting of two different social contexts related to the emotions of happiness and anger and two different discriminative stimuli for each social context, as well as animated peer and adult models. While preparing animations, attention was paid to selecting pastel and harmonious colours, choosing characters' clothes from daily life with simple colours and patterns, ensuring the background was free from distracting objects, and faithfully reflecting the social context in the scenario. Standardised facial expression templates were not used in the VAM videos; instead, the characters' facial expressions were manually drawn by the animator to reflect the emotions specified in the script in a natural and lifelike manner. According to this evaluation, the company that prepared animations made the necessary corrections. The opinions of three experts working in special education were received to evaluate the validity of the video images and the images with the animated model.

Experimental design

This study used the nested form of the non-concurrent multiple baseline design and adapted alternating treatments model across participants. The nonconcurrent multiple baseline design was used to reduce the time spent on baseline conditions or provide flexibility to include new participants when available [51]. Additionally, due to the need to address multiple research questions, the use of two independent variables, the implementation being carried out in the participants' homes, and the participants being available only after returning home from school, it was not possible to conduct sessions with all participants on the same day, and there were also time limitations. Therefore, this study was conducted using a replicated, nested form of the nonconcurrent multiple baseline design combined with an adapted alternating treatments design across participants. To ensure experimental control in this study using the non-concurrent multiple baseline model, (a) participants did not receive any training on the independent

Table 2. Materials used in probe, intervention, and generalization sessions.

Interventions	Emotions	Intervention sessions	Generalisation sessions
VM	Sadness	Biscuit, toy car	Exterior door of the house
	Fear	A mask of fear, puzzle, horn	Joke ring
VAM	Happiness	Football goal, ball, paper cup	Darts
	Anger	Telephone, chocolate	Toy ball

Note. VM: Video modelling; VAM: Video-animated modelling.

Table 3. Social scenarios and targeted social-affective behaviors.

Emotions	Social scenarios	Social-Affective Behaviours		
		Verbal expression	Facial expression	Gesture
Happiness	Scenario 1. A mother and a child enter the living room. The mother shows the child a ball and asks, "Shall we play a goal game?" The child expresses his willingness by nodding his head. The mother says, "Super, let's score a goal." The mother draws a line before the goal and puts the ball on the line. She points to the line and says, "Let us throw the ball into the goal." The child throws the ball towards the goal, and the ball enters the goal.	"I am happy." "Hurray! I did it! I am happy." "I did it, I'm happy." "Hurray, I am happy."	Laughing with the mouth open or closed	Raising both hands into the air, making fists
	Scenario 2. A mother and a child enter the living room. The mother asks, "Shall we play the game 'Tip the Glass Over'?" The child expresses his willingness by nodding his head. The mother says, "You are great; let's play." The mother arranges the glasses in a tower and draws a line in front of the glasses. She shows the line to the child and says, "Stop here." She gives the child a medium-sized plastic ball. The mother says, "Throw the ball into the glasses." The child throws the ball. The glasses tip over.	"I am happy." "Hurray! I did it! I am happy." "I did it; I'm happy." "Hooray, I am happy."	Laughing with the mouth open or closed	Raising both hands into the air, making fists
	Generalisation Scenario: Scenario 3. A father asks the child, "Shall we play the dart game?" by showing the dart to the child. The child expresses his willingness by nodding his head. The father says, "Okay, let's play." He draws a line in front of the dart and says, "Stop here." He gives the sticky ball to the child and says, "Throw it at the target." The child throws the sticky ball at the target, and the ball sticks to the dart.	"I am happy." "Hurray! I did it! I'm happy." "I did it. I am happy." "Hooray, I am happy."	Laughing with the mouth open or closed	Raising both hands into the air, making fists
Anger	Scenario 4. A mother and a child sit next to each other on the sofa. The mother asks the child, "Do you want to play a game on the phone?" by showing her phone to him. The child expresses his willingness by nodding his head. The mother hands the phone to the child but suddenly snatches the phone from him after the child plays on the phone for two minutes and pulls the phone towards her.	"I am very angry." "Give me the phone! I am very angry." "Give it to me. I'm so angry." "Give! I am so angry."	Frowning, closing the mouth, and pursing the lips.	Extending the hand to take the phone back and leaning the head forward (When frowning)
	Scenario 5. A mother and a child sit next to each other on the sofa. The mother asks the child, "Do you want chocolate?" The child expresses his willingness by nodding his head. The mother says, "Open your mouth," and brings the chocolate towards his mouth. When the child is about to bite the chocolate, the mother pulls the chocolate towards her. This process is repeated three times. The third time, the mother does not offer the chocolate again.	"I am very angry." "Give me the chocolate! I am very angry." "I will eat; give it to me. I'm so angry." "Give! I am so angry."	Frowning, closing the mouth, and pursing the lips.	Extending the hand to take the chocolate and leaning the head forward.
	Generalisation Scenario: Scenario 6. A father and a child sit opposite each other on the ground. The father asks the child, "Do you want to play with the car?" by pointing to the toy car in his hand. The child expresses his willingness by nodding his head. The father says, "Super, let us drive the car towards each other." The father sends the car to the child by driving it, and the child sends it back to his father. The process is repeated three times. The third time, when the car comes to the father, he takes the car and does not send it to the child again.	"I am very angry." "I will take the car." "Give! I am so angry." "Give! I am angry."	Frowning, closing the mouth, and pursing the lips.	Extending the hand to take the car and leaning the head forward.
Sadness	Scenario 7. A mother and a child sit next to each other on the sofa. The mother puts a plate in front of the child. She asks the child, "Do you want a biscuit/cake?" by showing the biscuit/cake in her hand. The child expresses his willingness by nodding his head. The mother opens the package and pours the broken biscuit pieces onto the plate.	"I am very sad." "Uh! This biscuit/cake is broken. I am very sad." "It is broken; I am so sad." "This is broken. I am so sad."	Curling down the lips and lowering the eyebrows.	Tilting the head sideways
	Scenario 8. A mother and a child sit side by side. The mother shows the car in her hand and asks the child, "Do you want to play?" The child expresses his willingness by nodding his head. The mother says, "Here, play." While the child is driving the car, the tire comes off.	"I am very sad." "Uh/this car is broken/ the tire is broken. I am very sad." "This tire is broken. I am so sad." "The car is broken." "I am very sad." "Uh/this door is broken. I am so sad." "The door is locked; I am so sad."	Curling down the lips and lowering the eyebrows.	Tilting the head sideways.
	Generalisation Scenario: Scenario 9. A father asks the child, "Let us go out. Shall we go for a walk?" The child nods, "Yes." The father says, "Great, come on, open the door." The child tries to open the door, but it does not open.	"I am very sad." "Uh/this door is broken. I am so sad." "The door is locked; I am so sad."	Curling down the lips and lowering the eyebrows.	Tilting the head sideways.

(Continued)

Table 3. Continued.

Emotions	Social scenarios	Social-Affective Behaviours		
		Verbal expression	Facial expression	Gesture
Fear	Scenario 10. A mother puts on a mask of fear and calls out to the child, "Come in." The child comes in and sees someone wearing the mask of fear.	"I was very scared." "Hihhh, I am so scared!" "Yaaa, I am so scared!" "I was so scared."	Opening the eyes and mouth, and raising the eyebrows.	Putting the head and face between the hands.
	Scenario 11. A mother and a child sit next to each other on the sofa. The mother asks the child, "Do you want to do puzzles?" The child expresses his willingness by nodding his head. The mother puts the puzzle in front of the child and says, "Here, play." When the child does the puzzle, she suddenly blows the horn in her hand.	"I was very scared." "Hihhh, I was so scared!" "Yaaa, I was so scared!" "I was so scared."	Opening the eyes and mouth, and raising the eyebrows.	Putting the head and face between the hands.
	Generalisation Scenario: Scenario 12. A father and a child sit next to each other on the sofa. The father puts a joke ring on his finger, holds it out to the child, and says, "Oh, there are fish swimming in this. Have a look." The child approaches the ring. The father squeezes the water-filled reservoir in the ring, and the water splashes on the child's face.	"I was very scared." "Hihhh, I was so scared!" "Yaaa, I was so scared!" "I was so scared."	Opening the eyes and mouth, and raising the eyebrows.	Putting the head and face between the hands.

variables, (b) reliability data were collected from 30% of all sessions, (c) parents, siblings, and teachers were informed not to teach the targeted behaviours, and (d) participants were selected from among those who volunteered to participate in the study, and one additional participant was included as a substitute. Moreover, to ensure experimental control in terms of the adapted alternating treatments model, emphasis was placed on selecting functionally similar but independent and equally difficult dependent variables for each independent variable to eliminate factors threatening internal validity. Additionally, factors threatening internal validity were controlled. These factors included history effects, participant attrition, maturation, testing effects, carryover effects, sequence effects, and instrumentation effects. To control for history effects, the participants' teachers were informed about the purpose of the study and were asked not to provide any instruction related to the targeted skills. To prevent participant attrition, efforts were made to maintain the motivation of the participating children and their families and to encourage regular participation in the study. Intervention sessions were conducted five to six days per week to prevent the intervention process from becoming prolonged, thereby minimising maturation effects. The study was designed according to a nonconcurrent multiple baseline model to reduce testing effects. To control for sequence effects, the independent variables were presented in an alternating manner. Carryover effects were minimised by providing at least a 45 min interval between the implementation of the independent variables. To control for instrumentation effects, treatment integrity and interobserver agreement data were collected. In addition, prior to the intervention, participants did not interact with the independent variables, thereby minimising measurement-independent variable interaction effects.

Dependent variables

The study's dependent variable was the performance of the participants who displayed appropriate SABs within 5 s after the emergence of a discriminative stimulus eliciting SABs in the form of verbal and facial expressions and gestures related to the emotions of happiness, sadness, anger, and fear, in line with the context (Please see details in Table 3). This study focused on the four basic emotions (happiness, sadness, anger, and fear), known to play a role in the comprehension and expression of emotions appropriate for the context [52]. The targeted emotions were selected based on relevant research (e.g., [3,19,53], as these emotions are believed to be the most common emotions pre-school children encounter in their daily peer interactions.

Discriminative stimuli

Scenarios describing social contexts were prepared, and the discriminative stimuli that caused the emergence of SABs related to emotions in these scenarios were determined. Discriminative stimuli were identified on the basis of objects, toys, and activities. Table 4 contains discriminative stimuli for probe, intervention, and generalisation sessions.

Table 4. Discriminative stimuli used in the intervention and generalization sessions according to emotions.

Emotions	Discriminative stimuli		
	Intervention sessions		Generalisation sessions
Happiness	Goal scoring	Glass tipping	Target shooting in darts
Anger	Picking up the phone while playing with it	Whether or not to take a favourite food to the mouth	Taking away a favourite toy
Sadness	Car breakdown while playing with the car	The favourite biscuit turns out to be broken	Failure to open the door when going out
Fear	Seeing someone wearing a mask of fear	Sudden loud honking of the horn while playing a puzzle game	The joke ring squirting water

Independent variables

The independent variables were the instructional practices with VM and VAM. The participants were shown videos on the tablet that used the peer model for VM and videos that used the animated model for VAM. Afterwards the participants were expected to perform the skills when they were guided to the setting where they would perform the scenarios.

Three expert opinions were sought regarding the difficulty level of behaviours identified among SABs. Based on difficulty levels, “happy” and “sad” were evaluated as being at the same level, and “fear” and “anger” were also evaluated as being at the same level. Accordingly, one emotion from the “happy-sad” category and one emotion from the other group (“fear-anger”) were randomly assigned to either the VAM or the VM.

General procedures

The general procedure consisted of probe (baseline, intermittent probe), intervention, maintenance, and generalisation sessions. All study sessions were conducted at the participants’ homes, five days a week, at times convenient for participants. Intervention sessions were organised as two sessions per day. Before the intervention sessions began, the order of the VM and VAM was determined at random, and thus counterbalanced across days for each participant. In each intervention session, the order of emotions was alternated.

Baseline and intermittent probe sessions

The baseline sessions were organised as one probe session per day for the participant to be instructed only, and a total of eight trials, four different contexts, and two different discriminative stimuli were presented in an unpredictable order. The single opportunity method was used for the participant to emit the SABs, including verbal expression, facial expression, and gesture as a whole. A response was scored as correct only when the participant emitted all three components (verbal expression, facial expression, and gesture) together. In such cases, a “+” was recorded in the relevant column of the data collection form, and the response was accepted as a correct response. If the participant displayed an incorrect response in one or more of the three components, a “–” was recorded in the relevant column of the data collection form, and the response was evaluated as an incorrect response. The baseline sessions were conducted as follows: The researcher prepared materials and arranged the setting prior to the sessions. She and the child entered the play area. The mother said, “We are going to play a game with you. Are you ready?” and provided an attention-grabbing cue to the child. When the child expressed his readiness, the mother verbally reinforced the child’s participation (e.g. “Well done”). She initiated interaction with the child regarding the game and then presented one of the discriminative stimuli (e.g. a puzzle for the child to play with, followed by a honking horn) that would elicit SABs. When the child responded correctly using verbal expressions (“I was very scared”), facial expressions (opening the eyes and mouth, raising the eyebrows), and gestures (putting the head or face between the two hands) related to the emotion of fear within five seconds, he was verbally reinforced (“Well done, superb!”). If the child did not respond appropriately using a verbal expression (“I was very scared”), facial expression (opening the eyes and mouth, raising the eyebrows), and gesture (putting the head or face between the two hands) or did not respond at all within five seconds, the researcher proceeded to the next trial. At the end of the session, the child’s participation was reinforced verbally (“You worked very well with me, well done”).

An intermittent probe session was held after two instruction sessions for each independent variable to determine the participants' performance in the target behaviours. The intermittent probe sessions continued until the participants gave 100% correct responses for three consecutive sessions. The same process was followed as in the baseline sessions.

Intervention sessions

The intervention sessions were organised at the participants' homes, with two sessions per day. In the sessions with VM, the researcher prepared materials and arranged settings in accordance with each scenario. The mother and child sat side by side on the sofa. The mother pointed to the video on the tablet and said, "Look, here are two people playing various games. Let us watch which games they are playing. Let's watch the video together," to draw the child's attention. The mother played the video and presented the target stimulus to the child, saying, "Watch the video." When the child did not watch the video, the mother drew the child's attention to the video by saying, "Watch the video." All the mothers performed the same actions and used the same expressions as in the scenarios they watched during intervention sessions. In one trial, the mother showed the video image of the discriminative stimulus for only one emotion. The mother verbally reinforced the child when he watched the video carefully. After watching the video, the mother and the child went to the play area. The mother said, "We are going to play the game you watched in the video. Are you ready?" to attract the participant's attention. After the child expressed his readiness, the mother initiated the interaction with the participant by presenting the discriminative stimulus related to the scenario (e.g. honking the horn a while after giving the child a puzzle to play). When the child responded correctly using verbal expressions (e.g. "I was very scared"), facial expressions (e.g. opening the eyes and mouth, raising the eyebrows), and gestures (e.g. putting the head or face between the two hands) related to the emotion of fear within five seconds and completed the skill within 10s, the researcher provided verbal and edible reinforcers. If the child did not respond appropriately to the emotion of fear within five seconds or did not respond at all, the researcher proceeded to the next trial. Then, the subsequent trial was initiated. The second trial was conducted by showing the second discriminative stimulus for the same emotion; the third trial was conducted by showing the first discriminative stimulus for the other emotion (sadness) (e.g. the favourite toy was broken). The fourth trial was conducted by showing the video image of the second discriminative stimulus for the other emotion (sadness) (e.g. the favourite biscuit was broken) and playing the game in the video. The order of the emotions was alternated. The inter-trial interval was set as five seconds. At the end of the intervention sessions, the children were verbally reinforced for cooperation ("You worked very well with me, well done").

Teaching with VAM was carried out in the same way as the VM intervention process. The difference was that images prepared with a video-animated model for the emotions of happiness and anger were shown, and games related to these emotions were played.

Maintenance and generalisation sessions

Maintenance sessions were held two, four, and six weeks after the criteria for the target behaviours were met. Pre-test-posttest generalisation sessions were conducted to determine whether participants could generalise the acquired behaviours to different people (father), settings (another room in the house), and scenarios (different social scenarios). The same process was followed in the maintenance and generalisation sessions as in the baseline sessions.

Prior to the generalisation sessions, fathers were informed about the social scenarios and the behaviours expected from them during the sessions. In addition, the implementer modelled the implementation steps for the fathers, including gaining the child's attention, presenting the discriminative stimulus, waiting for five seconds, reinforcing correct responses, and proceeding to the next trial following incorrect responses or the absence of a response.

Efficiency

To examine potential differences in efficiency between instructional interventions and to compare their relative efficiency, four types of efficiency data were collected, as recommended by Wolery et al. [54].

Accordingly, to compare the efficiency of the VM and VAM interventions, data were collected for each participant on (a) the number of instructional sessions required to meet the performance criterion, (b) the total number of trials conducted until criterion attainment, (c) the number and percentage of incorrect responses, and (d) total instructional time.

Effect size

Effectiveness data were collected by recording participants' correct and incorrect responses related to the target skills (SABs) during baseline, intermittent probe, maintenance, and generalisation sessions using standardised data collection forms. For each session, the percentage of correct responses was calculated. Effectiveness was evaluated through both visual (graphical) analysis and effect size estimation. To quantify the magnitude of the functional relation between the independent and dependent variables, the Tau-U effect size index was calculated. This index is commonly recommended for single-case experimental designs to supplement visual analysis and provide an objective estimate of intervention effects, while controlling for monotonic trends within phases [55].

Social validity

Semi-structured interviews were conducted with the participating children's mothers and teachers to assess the importance of the research purpose, methodology appropriateness, and the significance of the results. Ten open-ended questions were prepared for mothers and eight for teachers (Please see details in Table 5). The data obtained were analysed qualitatively.

Reliability

Inter-observer agreement (IOA) and treatment integrity data were collected in 30% of the sessions. Reliability data were collected by watching unbiased videos. The observers recorded the data obtained from the sessions on the relevant data collection forms, and these records were compared with the data forms completed by the observers. IOA was calculated based on the emission of all SAB components

Table 5. Social validity questions for mothers and teachers.

Mothers	Teachers
1. Do you think it is important to teach your child the behaviours of expressing emotions using verbal and facial expressions and gestures? Why do you think so? 2. Do you think learning to express emotions through verbal and facial expressions and gestures contributes to your child's life? Why do you think so? 3. Do you think that video modelling is effective in ensuring that your child learns to express emotions using verbal and facial expressions and gestures? Why do you think so? 4. Do you think teaching with video-animated modelling is effective in ensuring that your child learns to express emotions using verbal and facial expressions and gestures? Why do you think so? 5. After teaching your child to express emotions using verbal and facial expressions and gestures, did you observe your child displaying these behaviours in different social contexts and in the presence of different people? If yes, in which situations did you observe these behaviours? 6. Based on the video images you watched, which of the video modelling and video-animated modelling teaching practices do you think is more effective for your child in teaching expressing emotions using verbal expressions, facial expressions, and gestures? Why? 7. If this study were to be re-planned for teaching different emotions, which emotions would you prefer to be taught? Why? 8. If a new study were to be conducted, would you prefer video modelling or video-animated modelling for teaching different emotions? Why? 9. Can you explain what you like about this study in a few sentences? 10. If this study has aspects that you do not like, can you explain them in a few sentences?	1. Do you think teaching your students how to express emotions using verbal and facial expressions and gestures is critical? Why do you think so? 2. Based on the video images you watched, do you think your student learned to express emotions using verbal and facial expressions and gestures with video modelling and video-animated modelling? Why do you think so? 3. In the process of learning to express their emotions using verbal and facial expressions and gestures, did you observe that your student exhibited these skills in situations where they needed to display social-affective responses? If yes, what did you observe? 4. If you were to plan a new study on teaching different emotions, which emotions would you prefer to be taught? Why? 5. What do you think about using video modelling and video-animated modelling in your individual studies? 6. What do you think about using video modelling and video-animated modelling in inclusive settings? 7. Would you prefer to use video model or video-animated modelling more in an inclusive setting? Why? 8. What are your opinions on the study whose videos you watched?

together. Agreement was defined as both observers scoring the entire SAB stefan in the same way (correct or incorrect). IOA data were calculated using the formula "[Agreement/(Agreement+Disagreement)] × 100" [56]. IOA was calculated as 100% for all participants in the baseline, intervention, intermittent probe, maintenance, and generalisation sessions.

Treatment integrity data were calculated using the formula "Observed implementer behaviour/Planned implementer behaviour × 100" [57]. Treatment integrity data for the baseline, probe, maintenance, and generalisation sessions were collected based on the following behaviours: (a) preparing materials and arranging the setting, (b) going to the play area with the child, (c) drawing the child's attention by initiating interaction in an appropriate manner, (c) delivering the discriminative stimulus, (d) waiting 5s for the response, (e) responding appropriately to the child's responses, (f) having an interval between trials, and (g) reinforcing cooperation. Treatment integrity data for the intervention sessions of VM and VAM were collected based on the following behaviours: (a) preparing a video on the tablet, (b) preparing materials and arranging the setting, (c) drawing the child's attention to the video, (d) delivering the target stimulus (e.g. "Please watch the video."), (e) having the child watch the video, (f) reinforcing the child for watching the video, (g) going to the play area with the child, (h) drawing the child's attention by initiating interaction in an appropriate manner, (i) delivering the discriminative stimulus, (j) waiting 5s for the response, (k) giving appropriate responses to the child's responses, (l) having an interval between trials, and (m) reinforcing cooperation. The treatment integrity items were identical across VM and VAM conditions, with the only difference being the format of the instructional videos presented during the sessions. To minimise potential drift over time, mothers were informed about the implementation steps prior to the intervention sessions, and treatment integrity was monitored throughout the study using a data collection form. The treatment integrity for all children was calculated as 100% in the baseline, maintenance, and generalisation sessions and 96.3% in the intermittent probe sessions. The treatment integrity obtained for the intervention sessions with VM was 96.6%, and that for teaching with VAM was 96.6%.

Results

Comparison of effectiveness findings

Figure 1 shows the data obtained from all participants' baseline, intervention, maintenance, and generalisation phases regarding the SABs targeted to be taught with VM and VAM. Emir, Aslan, and Kerem did not give any correct response to SABs in the baseline phase. When the instruction with VM and VAM was implemented, there was a change in the trend and level of the participants' data related to the SABs targeted to be taught in the desired direction. In the intervention phase, all participants met the criterion by demonstrating the SABs taught with VM after ten, eight, and eight intervention sessions, respectively. Furthermore, all participants met the criterion by demonstrating the SABs taught with VAM after eight, ten, and eight intervention sessions, respectively, at a 100% accuracy level for three consecutive sessions. In the maintenance phase, all participants maintained the SABs taught with VM and VAM two, four, and six weeks after the intervention. In the generalisation pre-test session, none of the participants responded correctly to the SABs targeted to be taught with both VM and VAM. However, in the generalisation posttest session, they generalised SABs across different people (father), settings, and contexts with 100% accuracy.

Comparison of efficiency findings

When the efficiency of VM and VAM in teaching verbal and facial expressions and gestures as the targeted SABs was compared, the responses were evaluated together. Table 6 contains data on these variables for all participants for the three SABs as a whole. The findings varied when VM and VAM were compared in terms of efficiency. Table 6 shows that no differences were observed between the two instructional practices in terms of the number of trials and instructional sessions for Kerem. However, VAM was more efficient with respect to the total number of incorrect responses for Emir, whereas VM was more efficient in terms of total instructional time for Kerem. In addition, VM was more efficient than VAM across all efficiency parameters for Aslan.

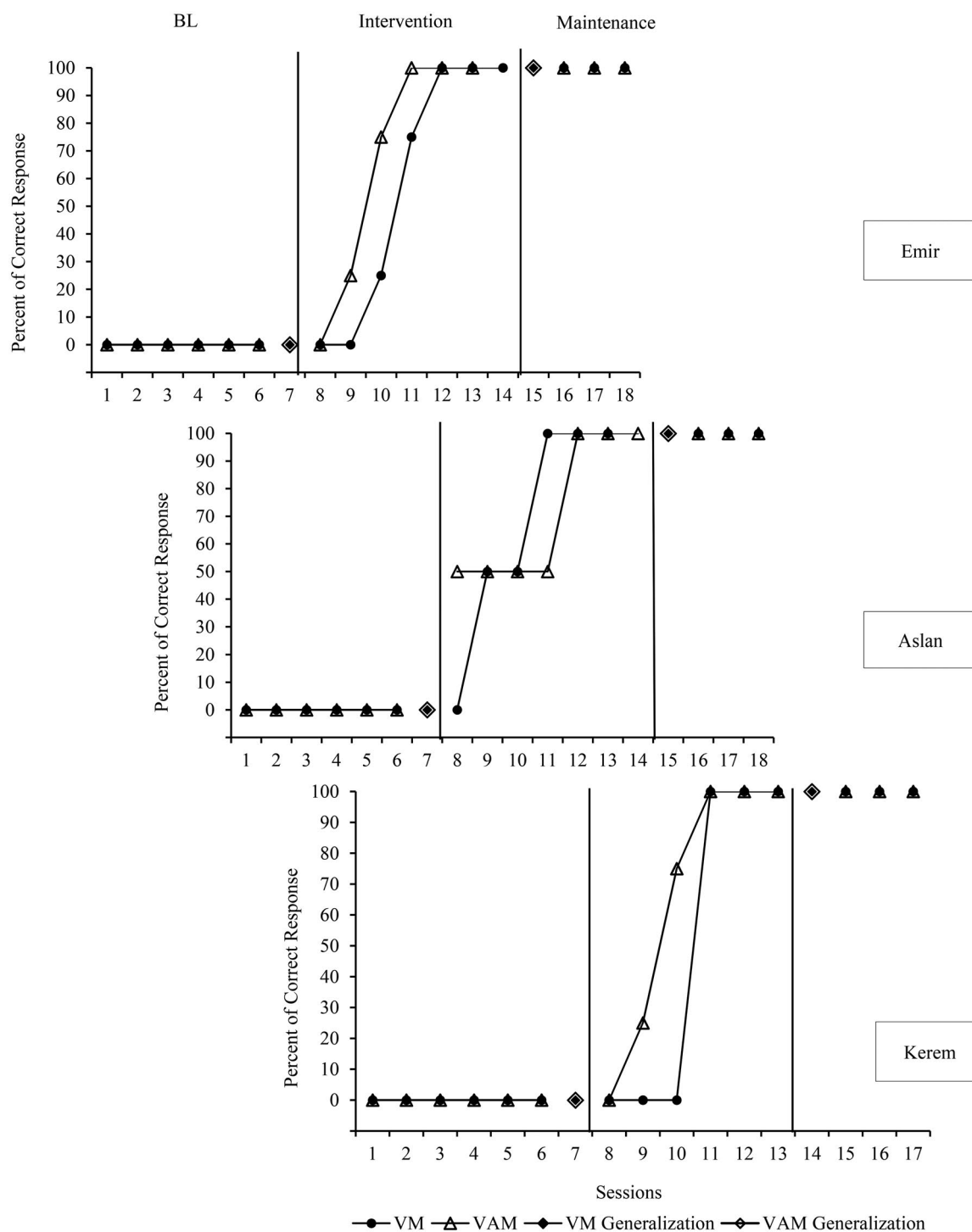


Figure 1. The percentage of correct responses of the participants on teaching social-affective behaviours in baseline, intervention, maintenance and generalisation sessions.

Comparison of effect size findings

To reveal the magnitude of the functional relationship between VM and VAM in teaching the components of SABs together, the effect size calculations for Tau-U were performed separately for each participant. Emir's findings for the effect size were 83.3% for VM and 90% for VAM. Aslan's findings for the effect size were 90% for VM and 100% for VAM. On the other hand, Kerem's effect size findings were 59% for VM and 90% for VAM. The average of the effect size calculations for all participants was 79% for VM and 93% for VAM.

Table 6. Efficacy data on teaching social-affective behaviors with VM and VAM.

Participants	Interventions	Number of intervention sessions	Number of trials in intervention sessions	Number and percentage of incorrect response		Duration of intervention sessions (h:m)
				N	%	
Emir	VM	10	40	30	75	38:52
	VAM	8	32	16	50	35:52
Aslan	VM	8	32	25	78.12	32:8
	VAM	10	40	27	67.5	54:54
Kerem	VM	8	32	31	96.87	29:45
	VAM	8	32	28	87.5	43:7

Note. VM: Video modelling; VAM: Video-animated modelling.

Social validation

Social validity findings showed that participants' mothers and teachers expressed positive opinions about the study's purpose, implementation, and findings. During the interviews with the mothers, they stated that the skills of expressing emotions acquired by their children effectively reduced problem behaviours, such as yelling and aggression, their children could generalise the skills they had learned to different social contexts, VM and VAM attracted their children's attention, they were happy to participate in such a study actively, and they would prefer the emotions of boredom and disgust to be taught if a similar study were planned.

During interviews, teachers emphasised the importance of children's ability to express their emotions for acceptance by their peers. They noted that children exhibited self-harming behaviours due to their inability to express emotions before the training. However, these problematic behaviours decreased after the training, as children learned to express their emotions. The teachers also stated that they found both teaching practices effective and that if a similar study were planned, they would prefer the skills of expressing oneself when in pain to be taught.

Discussion

The study examined whether VM and VAM differed in terms of effectiveness and efficiency in teaching SABs, including verbal and facial expressions and gestures. The effectiveness findings showed no difference between VM and VAM in terms of effectiveness in teaching three SABs to children with ASD as a whole. The literature review found only one study that compared VM with VAM in terms of functional relationship [41]. The current study's findings align with Li et al.'s [41] study, revealing that both VM and VAM effectively teach targeted skills, as evidenced by the calculated effect sizes. However, the current study and Li et al.'s study differ significantly in terms of methodology and findings. The methodological differences in Li et al.'s study can be listed as follows: (a) teaching communication and social initiation as social communication skills with predetermined lines in the context of opportunities created without emotion and without a scenario in a naturalistic setting (e.g. home, etc.), (b) preparing VM, VAM, and mixed VM (VM+VAM) using adult models and comparing the effectiveness of these interventions, (c) creating VAM by adding rabbit faces as animated characters to the face area of the adult models in VM only, (d) using the nested form of multiple baselines, alternating treatments, and multi-treatment models, and withdrawing the intervention by accepting the target skills as reversible behaviours between the intervention phases. In the current study, both interventions were effective in teaching SABs, with no consistent pattern of differences across participants. However, Li et al. reported VAM to be more effective than VM and mixed VM to be more effective than VM and VAM, which reveals the most fundamental difference between the two studies in terms of findings. Therefore, the current study potentially offers a novel perspective by examining differences between peer-modelled VM and 2D animated VAM for teaching verbal and facial expressions and gestures within the naturalistic socio-affective contexts.

A review of studies on teaching one, several, or all of the SABs shows that only two studies comparing the effectiveness of VM with a live model as a different teaching practice aimed to teach one (verbal expression) [32] or three (verbal and facial expression and intonation) of the SABs, including affective response [2]. Charlop-Christy et al. [32] reported that VM was more effective when they analysed data on verbal expression of emotions. However, the current study found VAM and VM to be effective in teaching all three behaviours, including verbal expressions. Gena et al. [2] reported that both VM and

live modelling effectively taught three behaviours as a whole. In line with these explanations, the current study contributes to the literature by comparing the effectiveness of VM and VAM in teaching three SABs as a whole. In the future, it may be recommended to conduct more studies in which one, several, or all of the SABs and different skills are compared by presenting VAM and VM.

Considering the studies that aimed to teach SABs to children with ASD through VM, the previous studies differ from the current study in the following aspects: (a) teaching one or two of the SABs (e.g. [29,30]), (b) conducting research in an institute within the school or university (e.g., [58,59]), (d) including primary school children (e.g., [33]), and (e) VM being accompanied by various teaching practices (e.g., [60]). Considering all differences together, the current study obtained important findings regarding the effects of VM in teaching verbal expressions, facial expressions, and gestures together with children's mothers at home, which is the child's naturalistic setting after social contexts are created. In the future, it may be useful to conduct more research on teaching all SABs together in social contexts with VM or different teaching practices.

Few studies have demonstrated the effectiveness of VAM in teaching different skills [41,43]. However, no study determined the effectiveness of teaching SABs to children with ASD through VAM. The current study's findings are consistent with those of previous studies, showing the effectiveness of VAM in teaching different skills. In the future, more research can be conducted to determine the effectiveness of teaching various skills and SABs separately or as a whole with VAM.

The study's findings show that VM and VAM were effective in ensuring that the participating children maintained the acquired SABs two, four, and six weeks after the intervention. In the study by Li et al. [41], maintenance data were collected, but the videos were not withdrawn during the maintenance sessions, and participants continued to watch the videos. In contrast to Li et al.'s study, the current study followed the same process as the probe sessions during the maintenance sessions, and no videos were shown to the participants. The effectiveness findings of the current study, indicating that the children maintained the three SABs two, four, and six weeks after the intervention with VM, are similar to studies where skills were maintained from one week to six months (e.g., [32,58,59]). However, they differ from a study in which the acquired skills were not maintained after four weeks [22] and maintenance data were not collected [4]. In the future, it can be recommended that the number of studies on the maintenance of SABs be increased using different teaching practices, such as VM and VAM.

The study's findings show no difference between VM and VAM in terms of effectiveness in generalising three SABs across different people (mothers), settings (a different room in the house), materials, and contexts. This finding coincides with the findings of Li et al. [41], indicating that VM and VAM are effective in terms of generalisation across materials. The current study's findings support the studies investigating the effects of VM on the generalisation of SABs (e.g., [2,32,58]). The present work expands the limited literature by generalising the acquired skills to contexts different from those in teaching, utilising distinctive stimuli, a different person, and a different setting. Therefore, it may be recommended that more generalisation studies be conducted in different social contexts created using various materials with familiar and unfamiliar people and in naturalistic settings.

The findings varied when the efficiency of VM and VAM was compared. VAM was more efficient than VM in terms of the number of incorrect responses for Emir and Kerem. However, VM was more efficient in terms of total instructional time for Aslan and Kerem. Li et al. [41] and Gena et al. [2] did not collect efficiency data. On the other hand, Charlop-Christy et al. [32] reported that VM was more time- and cost-efficient in teaching verbal expressions. In the future, more studies should be conducted to clarify the efficiency of VM and VAM in teaching skills.

The Tau-U effect size findings showed that VM was highly effective for Emir and Aslan and moderately effective for Kerem. VAM was effective for Emir and Kerem and very effective for Aslan. When the components were examined separately in terms of effect size, it was seen that VM and VAM were very effective for all participants in teaching verbal expressions and gesture behaviours. The finding of Li et al.'s [41] study, in which VM and VAM effectively taught social communication skills according to Tau-BC calculations, differs from the current study's findings in that VAM was more effective when compared to VM, and mixed VM was more effective when compared to VM and VAM. The other comparison studies by Charlop-Christy et al. [32] and Gena et al. [2] did not calculate the effect size of interventions in teaching one or more of the SABs. In the future, it may be recommended to perform effect size calculations in studies to determine which intervention is more effective.

The social validity findings obtained through semi-structured interviews show that the participating children's mothers and teachers expressed positive opinions about the study's importance, implementation, and the effectiveness of the methods. The study conducted by Li et al. [41] also overlaps with the current study regarding collecting social validity data. However, it differs from the current study in terms of collecting data through a questionnaire. Furthermore, it is noteworthy that the studies comparing the effectiveness of VM and live models [2,32] did not collect social validity data. It would be beneficial for future studies to collect social validity data.

Although the study's effectiveness, generalisation, and maintenance findings are positive, some points during the intervention process should be discussed. First, in the context This can be explained by the fact that Emir imitated the facial expressions of his parents when the ball entered the goal because he watched matches with them. Second, Kerem made progress in the third probe session with VAM and the fourth probe session with VM. This was associated with his focus on eating the food offered, especially in the contexts of "broken biscuit" and "not giving his favourite food." In the context of "the toy broke down," he focused on fixing the car when the car's wheels came off and did not exhibit the expected behaviour. Therefore, biscuits and chocolate, which Kerem did not prefer to eat, were identified and adapted by attaching knitted tires to the places where the wheels came off. Additionally, a different mask was used in the contexts where the mask of fear was used because it lost its frightening effect for Kerem. The targeted criterion was reached in the probe sessions organised after these adaptations.

The current study has some limitations. Experimental and logical analyses are recommended to ensure that dependent variables are equally difficult [61]. The fact that logical analysis was conducted and experimental analysis was not included in the study can be considered a limitation. Additionally, the limited number of participants inherent in single-subject research designs, along with the fact that all participants in the present study were male, may be considered a limitation in terms of the generalisability of the findings. It is also a limitation that generalisation sessions were organised at the participants' homes and not conducted in social settings, where social contexts often occur. The fact that the study targeted only basic emotions and did not target intonation, one of the SABs, can also be considered a limitation. Finally, while creating videos with the animated model, the fact that the program only allowed the preparation of two-dimensional animations limited the model's movements and caused the movements to be displayed piece by piece and to lose their naturalness.

For further research, it may be suggested that research aiming to teach SABs, including complex emotions in different social settings and naturalistic contexts, can be conducted. The effectiveness of VAM, which consists of three-dimensional animations, in teaching SABs can also be investigated. Furthermore, replication studies conducted with different age groups, developmental characteristics, and cultural contexts may contribute to understanding the generalisability of the findings. Future studies may also compare the effectiveness of VM and VAM in teaching different skill areas, such as social communication, play, daily living, and academic skills. In recent years, computer-based games (e.g. JeStiMule, FaceMaze, The Transporters, Emotiplay, ALTRIRAS, Zirkus Empathico) have frequently been preferred and found effective in studies targeting the teaching of one, several, or all components of SABs [62–66]. Therefore, future studies could examine the differences in effectiveness and efficiency between these games and VM or VAM in the teaching of SABs.

In conclusion, the findings indicated that VM and VAM were equally effective in teaching SABs, and neither intervention demonstrated a consistent superiority pattern across participants in terms of effectiveness. However, individual variability was observed in terms of effect size and efficiency outcomes. While VAM was found to be more efficient for some participants in terms of the total number of incorrect responses, VM appeared to be more efficient for one participant in terms of total instructional time. These differences may be related to the characteristics of the interventions. For some participants, the attention-grabbing nature of the animated model may have increased attention and reduced the number of errors, whereas for another participant, the human model may have facilitated imitation and faster performance due to being perceived as more natural and socially familiar. From a practical perspective, the findings suggest that the selection of an intervention should not depend solely on effectiveness, but also on variables such as child preference, preparation time, and accessibility. However, because the development of VAM materials requires additional time, technical support, and financial resources, practitioners may prefer VM in situations where comparable learning outcomes are obtained and ease of preparation and accessibility are prioritised.

Ethical statement

I confirm that all the research meets ethical guidelines and adheres to the legal requirements of the study country. Ethical committee decision information is as follows: (a) name of the committee: A University Social Sciences and Humanities Scientific Publication Ethics Committee, (b) the date of the decision information: 03.11.2020 and (c) the protocol number: 58192.

Author contributions

CRedit: **Selin Gokce**: Conceptualization, Formal analysis, Investigation, Validation, Visualization, Writing – original draft; **Burcu Ulke-Kurkcuoglu**: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

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